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Web Application Project Development Using Python & Django

FitAI: AI-Driven Fitness and Nutrition Tracking Web Application

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Submitted By

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CHAPTER 1: INTRODUCTION TO PROJECT

The increasing use of technology in daily life has significantly influenced how individuals manage their health and fitness. Despite the availability of numerous fitness applications, many users still struggle to maintain a balanced lifestyle due to lack of proper guidance, inconsistency in tracking activities, and absence of personalized recommendations.

Most existing systems focus only on specific aspects such as calorie tracking or exercise monitoring, without providing an integrated platform that combines both functionalities effectively. Additionally, these systems often lack intelligent insights that can help users make informed decisions about their diet and physical activities.

To overcome these challenges, the **FitAI** project is proposed as an advanced fitness and nutrition tracking web application. The system enables users to track their daily food intake, monitor exercise routines, and analyse calorie consumption and expenditure. It also provides personalized diet and workout plans based on user-specific data such as age, weight, height, and fitness goals.

The application is developed using **Django and Django REST Framework**, ensuring a scalable and modular backend architecture. The system uses **PostgreSQL** for efficient data management and implements **JWT-based authentication** for secure user access. Additionally, API documentation is supported using Swagger, making the system easier to test and integrate.

The FitAI platform is designed to evolve into an intelligent health management system by incorporating AI-based recommendations, predictive analytics, and integration with mobile applications and wearable devices in future phases.

CHAPTER 2: MAJOR COMPONENTS OF PROJECT

2.1: OBJECTIVES

The primary objectives of the FitAI project are to develop a comprehensive and intelligent system for managing user fitness and health data. The key objectives are as follows:

- To design and develop a web-based application for tracking diet and exercise activities
- To calculate and monitor daily calorie intake and calories burned by users
- To provide personalized diet and exercise plans based on user fitness goals
- To store and manage user health-related data such as age, height, weight, and medical conditions
- To generate analytical insights including daily, weekly, and monthly progress reports
- To implement an intelligent system for suggesting improvements in user diet and workout routines
- To build a scalable backend using Django REST Framework for future web and mobile integration

2.2: PROBLEM STATEMENT

In today's digital era, maintaining a healthy lifestyle has become increasingly difficult due to sedentary habits, irregular eating patterns, and lack of awareness about proper fitness practices. Although several fitness applications are available, they often provide limited functionality and fail to deliver a comprehensive solution for managing both diet and exercise.

Users commonly face challenges such as:

- Difficulty in accurately tracking daily calorie intake and calories burned
- Lack of personalized diet and exercise recommendations
- Inconsistent tracking of fitness activities
- Limited understanding of nutritional values of food
- Absence of meaningful insights to monitor long-term progress

Most existing systems are either fragmented or overly generic, offering one-size-fits-all solutions that do not consider individual user needs and health conditions. Additionally, many applications lack intelligent features that can adapt to user behaviour and provide actionable guidance.

Therefore, there is a need for an integrated and intelligent system that not only tracks user activities but also provides personalized recommendations, analytical insights, and a structured approach to achieving fitness goals effectively.

2.3: SOLUTION

To address the identified challenges, the **FitAI** system proposes a comprehensive and integrated web-based solution for managing fitness and nutrition effectively. The system combines diet tracking, exercise monitoring, and intelligent recommendations into a single platform.

The application allows users to register and create a personalized profile by entering essential health details such as age, height, weight, medical conditions, and fitness goals. Based on this information, the system provides customized diet and exercise plans tailored to individual needs.

Users can manually log their daily food intake and physical activities. The system automatically calculates calorie consumption and calories burned, helping users maintain a proper balance according to their fitness objectives. It also provides insights such as daily summaries and progress tracking.

Additionally, the system includes intelligent logic to assist users in decision-making. For example, if a user logs only diet data, the system can suggest appropriate exercises, and vice versa. This ensures flexibility while still maintaining guidance.

The backend is developed using Django and Django REST Framework, ensuring scalability and modularity. PostgreSQL is used for efficient data management, and JWT authentication provides secure access to the system. Swagger is integrated for API documentation and testing.

Overall, the FitAI system offers a structured, user-friendly, and scalable solution that helps users track, analyse, and improve their fitness journey effectively.

CHAPTER 3: METHODOLOGY ADOPTED TO CARRY OUT A PROJECT / INTERNSHIP

The development of the FitAI system follows a structured and systematic methodology based on the **Django MVT (Model–View–Template) architecture** combined with **REST API design principles**. This approach ensures modularity, scalability, and maintainability of the application.

Development Approach

The project is developed using an incremental and modular approach, where each component of the system is designed, implemented, and tested in phases. This helps in managing complexity and ensures that each feature is properly validated before moving to the next stage.

Project Phases

Phase 1: Project Setup and Core Development

- Creation of Django project and applications
- Setup of virtual environment
- Configuration of PostgreSQL database
- Implementation of core models (UserProfile, Food, Exercise, Logs, Plans)
- Setup of JWT-based authentication system

Phase 2: API Development

- Integration of Django REST Framework
- Creation of serializers for all models
- Development of RESTful APIs for:
 - ✓ User authentication
 - ✓ Profile management
 - ✓ Diet tracking
 - ✓ Exercise tracking
- Integration of Swagger for API documentation

Phase 3: Tracking and Analytics

- Implementation of meal and workout logging
- Calculation of calories consumed and burned
- Development of analytics features:
 - ✓ Daily reports
 - ✓ Weekly summaries
 - ✓ Goal tracking

Phase 4: Intelligent Recommendation System

- Implementation of BMI and BMR calculation logic
- Automated diet and exercise plan generation
- Smart suggestions based on user history
- Basic predictive analysis for calorie requirements

Phase 5: Future Integration (Planned)

- Development of frontend using React/Next.js
- Integration with mobile applications

- Support for wearable devices

System Architecture

The system follows the Django MVT architecture:

- **Model:** Defines the database structure and relationships
- **View:** Handles business logic and API responses
- **Template:** Used for frontend (initial stage)
- **REST APIs:** Enable communication between backend and frontend

Development Principles

- Modular and scalable design
- Secure authentication using JWT
- RESTful API standards
- Clean and maintainable code structure

CHAPTER 4: TOOLS AND TECHNOLOGY USED

The development of the FitAI project utilizes a combination of modern tools and technologies to ensure efficiency, scalability, and maintainability of the system.

Programming Language

- Python
It is used as the primary programming language due to its simplicity, readability, and strong support for web development and data processing.

Backend Framework

- Django
It is used as the core web framework for building the application. It provides built-in features such as ORM, authentication, and admin panel, which simplify development.
- Django REST Framework (DRF)
DRF is used to build RESTful APIs, enabling communication between backend and frontend systems.

Database

- PostgreSQL
It is used as the primary database for storing application data such as user profiles, food logs, exercise logs, and plans. It is chosen for its reliability, scalability, and support for complex queries.

Authentication

- JWT (JSON Web Token) – SimpleJWT
JWT is used for secure authentication. It allows stateless communication between client and server by generating access and refresh tokens.

API Documentation

- Swagger
Swagger is used to automatically generate API documentation. It helps in testing and understanding API endpoints.

Frontend Technologies

- HTML, CSS, Bootstrap
Used for basic user interface design.
- React / Next.js (Planned)
Will be used for building a modern and dynamic frontend interface.

Development Tools

- Visual Studio Code (VS Code)
Used as the primary code editor.
- Git
Used for version control and project management.

- Postman
Used for testing APIs during development.

Additional Libraries

- psycopg2 (PostgreSQL adapter for Django)
- Django Filters (for search and filtering functionality)

CHAPTER 5: DATA ON THE PROJECT

The FitAI system is divided into multiple modules, each responsible for handling specific functionalities of the application. This modular approach ensures better organization, scalability, and maintainability.

5.1: USER MANAGEMENT MODULE

This module manages user authentication and profile data.

- User registration and login using JWT authentication
- Profile creation and update
- Storage of user health data such as age, height, weight, and fitness goals

5.2: DIET TRACKING MODULE

This module allows users to track their food intake.

- Add and manage food items
- Log daily meals
- Calculate calorie intake and nutritional values (protein, carbohydrates, fats)

5.3: EXERCISE TRACKING MODULE

This module records physical activities performed by users.

- Add and manage exercises
- Log workout sessions
- Calculate calories burned based on activity duration

5.4: PLAN GENERATION MODULE

This module generates personalized plans for users.

- Create diet plans based on calorie requirements
- Generate exercise plans based on fitness goals
- Allow manual and automatic plan customization

5.5: ANALYTICS AND INSIGHTS MODULE

This module provides useful insights and reports.

- Daily, weekly, and monthly summaries
- Calorie balance tracking (consumed vs burned)
- Progress tracking and streak analysis

5.6: RECOMMENDATION MODULE

This module provides intelligent suggestions.

- Suggest diet and exercise improvements
- Analyse user behaviour and habits
- Generate smart recommendations based on past data

5.7: DATABASE MANAGEMENT

The system uses a PostgreSQL database to store and manage data efficiently.

Main Tables

- User
- UserProfile
- FoodItem
- MealLog
- Exercise
- ExerciseLog
- DietPlan
- ExercisePlan

Relationships

- One-to-One: User → UserProfile
- One-to-Many: User → MealLog, ExerciseLog
- Foreign Key relationships between logs and respective entities

The database is designed using normalization techniques to avoid redundancy and ensure data integrity.

5.8: SYSTEM FLOWCHART

Below is the logical flow of the system:

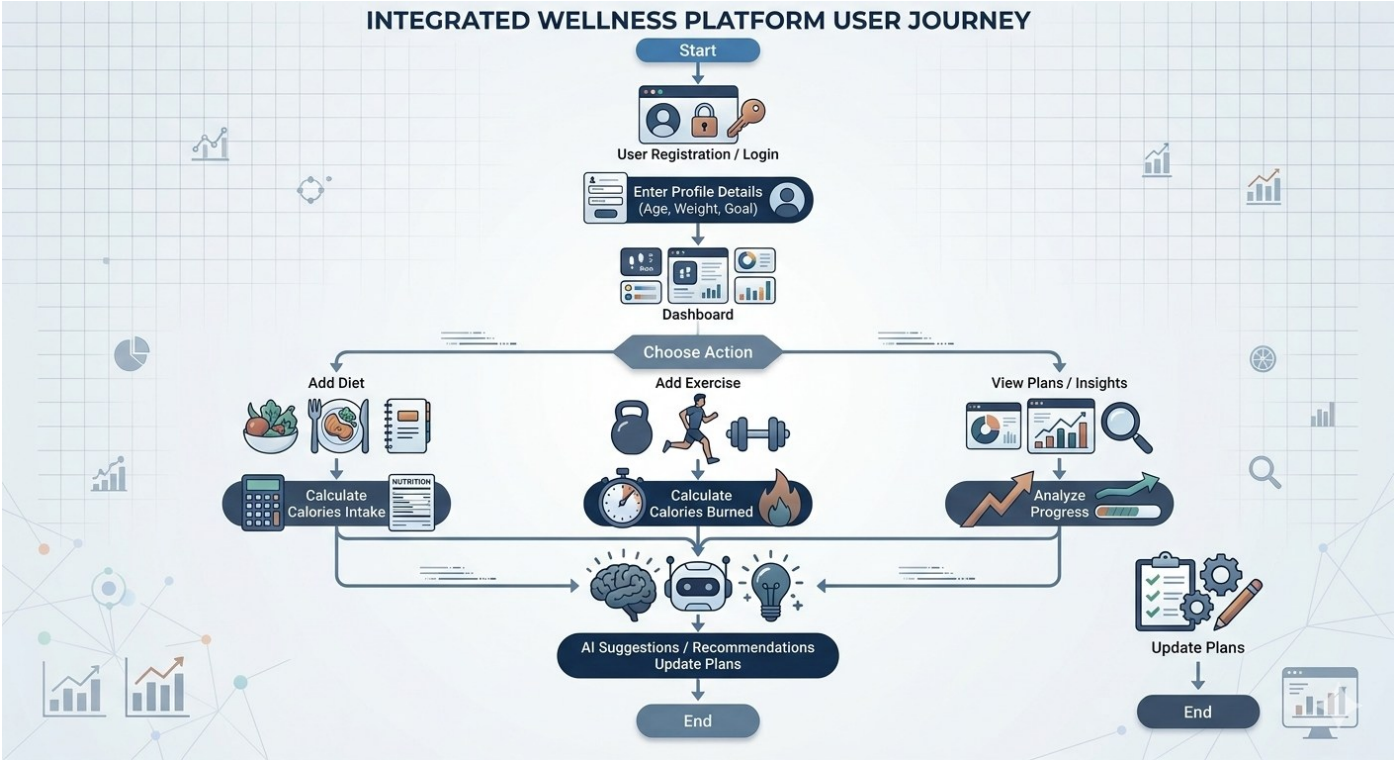


Fig 5.8.1: Flowchart

CHAPTER 6: OBSERVATION

6.1: EXPECTED OUTPUT

The FitAI system is expected to produce accurate and meaningful outputs that help users manage their fitness effectively.

- Successful user registration and secure login using JWT authentication
- Creation and updating of user fitness profiles
- Accurate calculation of daily calorie intake based on food logs
- Accurate calculation of calories burned based on exercise logs
- Generation of personalized diet and exercise plans
- Display of daily, weekly, and monthly fitness insights
- Proper functioning of REST APIs with Swagger documentation
- Efficient data storage and retrieval using PostgreSQL

The system should provide a smooth and user-friendly experience while ensuring correct calculations and reliable data handling.

6.2: SYSTEM SNAPSHOTS

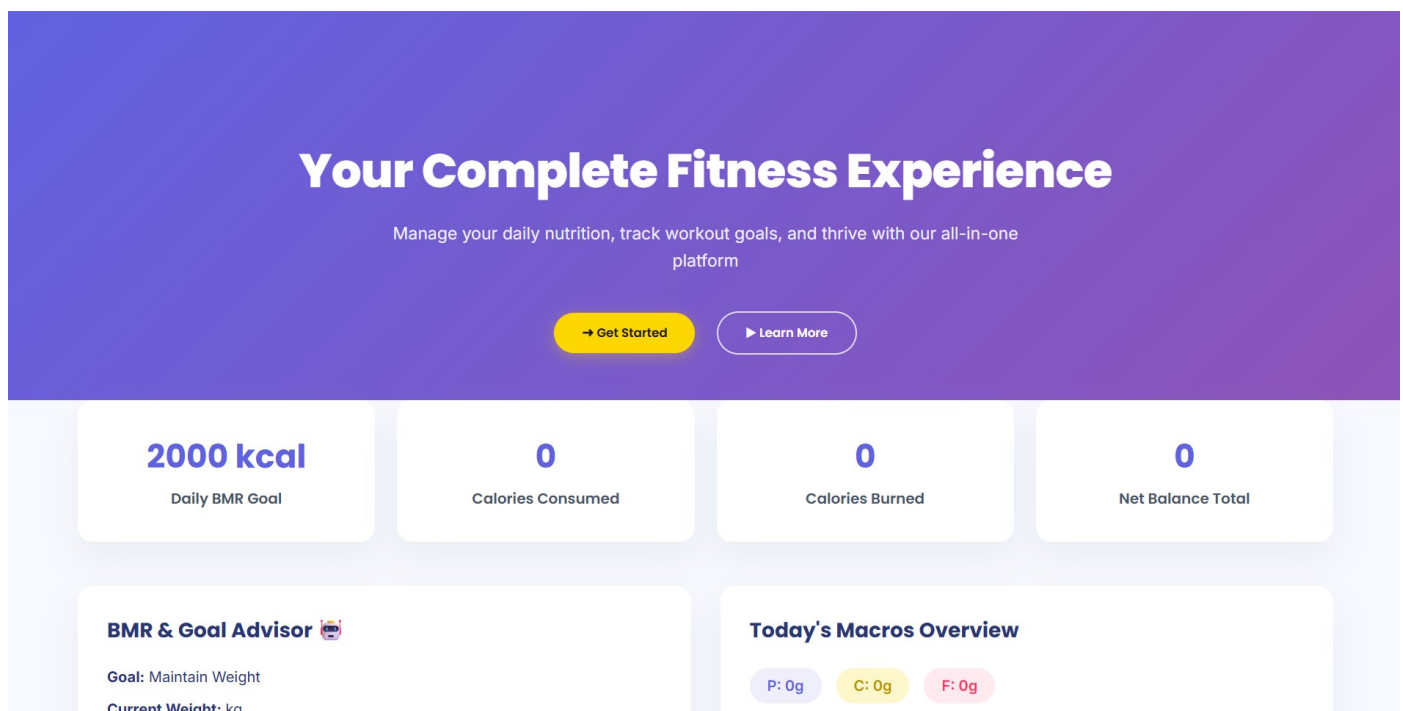


Fig 6.2.1: Dashboard

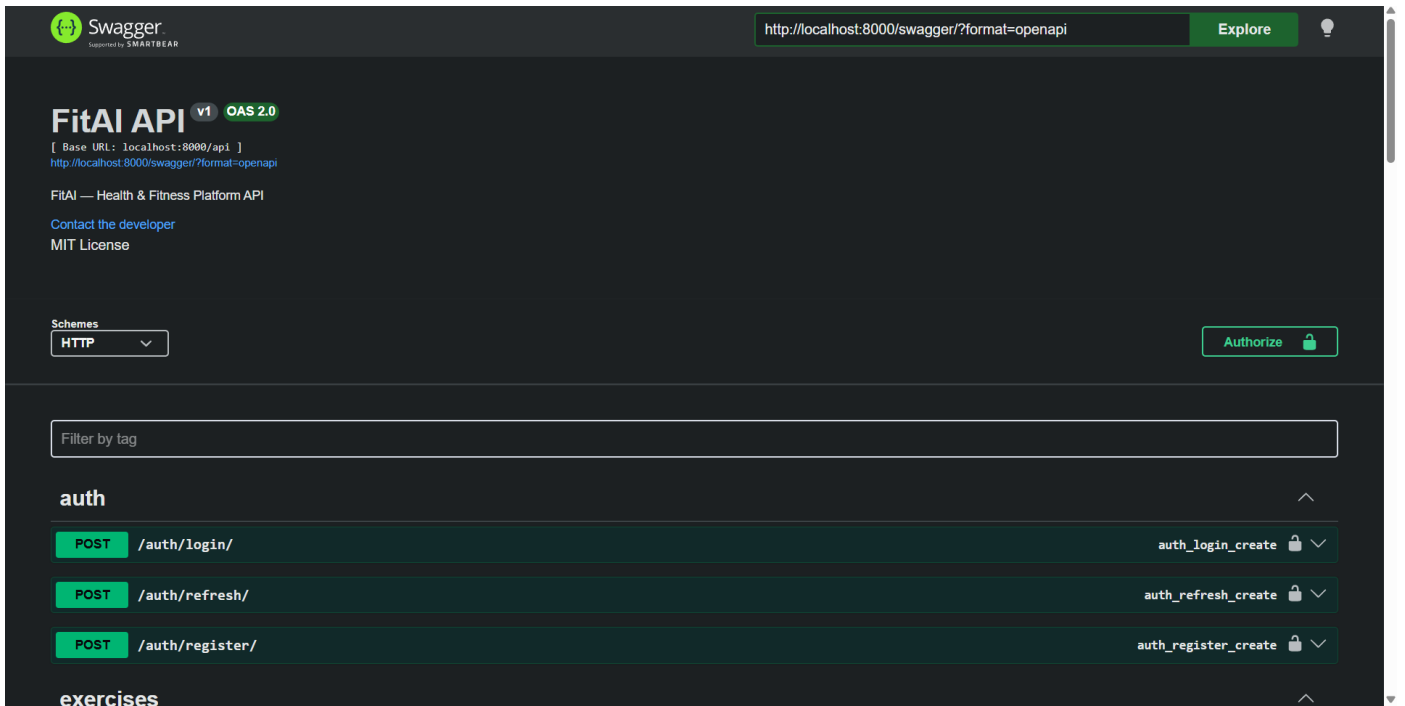


Fig 6.2.2: Swagger APIs

CHAPTER 7: FUTURE SCOPE

The FitAI system has strong potential for further development and enhancement. The following improvements can be implemented in future versions to make the system more advanced and user-friendly:

7.1: AI AND MACHINE LEARNING ENHANCEMENTS

- Implementation of advanced AI models for personalized diet and exercise recommendations
- Predictive analysis for calorie requirements and weight trends
- Intelligent habit detection based on user activity patterns
- Adaptive fitness plans that adjust automatically according to user progress

7.2: MOBILE APPLICATION DEVELOPMENT

- Development of Android and iOS mobile applications
- Real-time synchronization between web and mobile platforms
- Improved accessibility and user experience through mobile interfaces

7.3: WEARABLE DEVICE INTEGRATION

- Integration with fitness devices such as smartwatches and fitness bands
- Automatic tracking of steps, heart rate, and calories burned
- Real-time health monitoring and data synchronization

7.4: ADVANCED ANALYTICS AND VISUALIZATION

- Interactive dashboards with graphical representations (charts and graphs)
- Detailed performance analysis and historical comparisons
- Goal-based progress visualization

7.5: NOTIFICATION AND REMINDER SYSTEM

- Push notifications for workout and meal reminders
- Alerts for missed activities or exceeding calorie limits
- Personalized reminders based on user behavior

7.6: EXTERNAL API INTEGRATION

- Integration with global food databases (e.g., OpenFoodFacts)
- Access to a wide variety of nutritional data
- Automatic food recognition and suggestions

CHAPTER 8: CONCLUSION

The FitAI project successfully presents a comprehensive solution for managing fitness and nutrition through a web-based platform. The system integrates key functionalities such as diet tracking, exercise monitoring, calorie calculation, and personalized plan generation into a single unified application.

By utilizing modern technologies like Django, Django REST Framework, PostgreSQL, and JWT authentication, the system ensures scalability, security, and efficient data management. The modular design of the application allows easy expansion and integration with future technologies such as AI-based recommendations, mobile applications, and wearable devices.

The platform not only helps users track their daily activities but also provides meaningful insights that assist in making informed decisions regarding their health. This contributes to improved lifestyle management and encourages users to maintain consistency in achieving their fitness goals.

Overall, FitAI serves as a strong foundation for a smart and intelligent fitness management system, with the potential to evolve into a fully automated, AI-driven health assistant in the future.